



A Disk Rolling on a Horizontal Surface Without Slip

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(Received March 2002; accepted April 2002)

Abstract—A circular disk rolling on a horizontal surface without slip is a common example of a nonholonomic problem of Lagrangian mechanics. In the literature, very few such problems have been attacked developing simple and interesting results. Here, it will be shown first that it is possible to find a steady motion of the system, which is rolling in a circle. Then it will be shown that it is possible to find the expression for the frequency of small oscillations about this steady motion, including an approximate expression for the speed required for stability of that steady motion, which is accurate for circles large in radius compared to the disk. The solution is shown to reduce to the well-known critical speed for a disk rolling in a straight line. It is then surprising to note that the critical speed for stability is slower for motion in a circle than for the straight line. This result will be developed entirely analytically which seems to be quite unusual for such a nonholonomic problem of this complexity. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords—Dynamics, Lagrangian, Non-holonomic, Steady motion, Stability, Oscillations.

THE LAGRANGIAN APPROACH

The rolling disk is shown in Figure 1 in order to illustrate the generalized coordinates used here to establish its position. The five coordinates selected are X, Y, θ, ϕ, ψ , where X and Y locate the center of the disk projected vertically on to the horizontal plane and the three angles describe the angular position of the rigid body in an Eulerian fashion. The principal axes of moment of inertia are also identified with x as the axis of symmetry with the other orthogonal pair y and z with y not rotating with the body but remaining parallel to the horizontal plane. The rotation rate vectors are also shown on this figure.

For the Lagrangian approach there will be five Lagrange's equations, one for each coordinate required and two equations of constraint representing the condition of no slip on the horizontal plane.

The authors acknowledge the motivation provided by J. Ferry and E. R. Brandon to bring this work to completion. The first author also wishes to make special note of the extensive algebraic efforts of the second author that made and assured the accuracy of the analysis in this work. In addition, the editorial assistance of H. Tada is gratefully acknowledged.

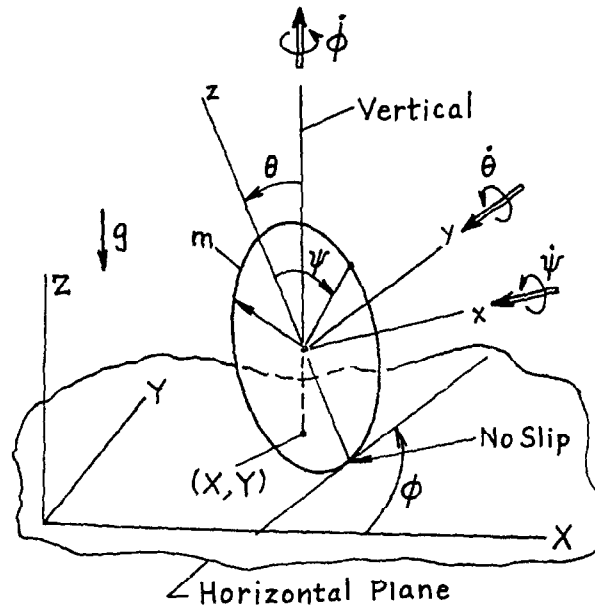


Figure 1.

The two constraint relationships may be expressed as

$$\begin{aligned}\delta X &= R \cos \phi \delta \psi - R \sin \theta \cos \phi \delta \phi - R \cos \theta \sin \phi \delta \theta, \\ \delta Y &= R \sin \phi \delta \psi - R \sin \theta \sin \phi \delta \phi + R \cos \theta \cos \phi \delta \theta.\end{aligned}\quad (1)$$

For the purpose of writing the kinetic energy, the angular velocities about the principal axes must be expressed in terms of the time rates of change of the selected coordinates, which results in

$$\begin{aligned}\omega_x &= -\dot{\psi} + \dot{\phi} \sin \theta, \\ \omega_y &= -\dot{\theta}, \\ \omega_z &= \dot{\phi} \cos \theta.\end{aligned}\quad (2)$$

Then the kinetic energy of motion may be expressed using the body axes rotation rates in the form

$$T = \frac{1}{2} m \bar{v}^2 + \frac{1}{2} [I_A \omega_x^2 + I_T (\omega_y^2 + \omega_z^2)], \quad (3)$$

where $\bar{v}$ is the velocity of the mass center and I_A and I_T are the axial (x -axis) and transverse axes moments of inertia. The potential due to gravity, g , can be written as

$$V = +mgR \cos \theta. \quad (4)$$

Then equations (2)-(4) may be combined to form the Lagrangian function, which is

$$\begin{aligned}L = T - V &= \frac{1}{2} m [\dot{X}^2 + \dot{Y}^2 + R^2 \sin^2 \theta \dot{\theta}^2] \\ &+ \frac{1}{2} \left[I_A (\dot{\psi} - \dot{\phi} \sin \theta)^2 + I_T (\dot{\theta}^2 + \dot{\phi}^2 \cos^2 \theta) \right] - mgR \cos \theta,\end{aligned}\quad (5)$$

which for each coordinate, q_r , obeys a Lagrange's equation for nonholonomic systems of the type

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_r} \right) - \frac{\partial L}{\partial q_r} = Q'_r, \quad (6)$$

where the Q'_r are the constraint forces, which with no slip do no work. This is expressed for a virtual displacement, δq_r , of the coordinates using the constraint relations, equations (1), each multiplied by the unknown Lagrangian multipliers, λ_n , which then gives

$$\begin{aligned} \delta(\text{work}) = 0 &= Q'_X \delta X + Q'_Y \delta Y + Q'_\theta \delta \theta + Q'_\phi \delta \phi + Q'_\psi \delta \psi \\ &= \lambda_1 \left(\begin{array}{c} \delta X + R \cos \theta \sin \phi \delta \theta \\ + R \sin \theta \cos \phi \delta \phi - R \cos \phi \delta \psi \end{array} \right) + \lambda_2 \left(\begin{array}{c} \delta Y - R \cos \theta \cos \phi \delta \theta \\ + R \sin \theta \sin \phi \delta \phi - R \sin \phi \delta \psi \end{array} \right). \end{aligned} \quad (7)$$

When the coefficients are matched in equation (7), Q'_r are determined as follows:

$$\begin{aligned} Q'_X &= \lambda_1, \\ Q'_Y &= \lambda_2, \\ Q'_\theta &= \lambda_1 R \cos \theta \sin \phi - \lambda_2 R \cos \theta \cos \phi, \\ Q'_\phi &= \lambda_1 R \sin \theta \cos \phi + \lambda_2 R \sin \theta \sin \phi, \\ Q'_\psi &= -\lambda_1 R \cos \phi - \lambda_2 R \sin \phi. \end{aligned} \quad (8)$$

Putting equations (8) and (5) into equation (6) gives the full Lagrange's equations

$$\begin{aligned} \frac{d}{dt} (m\dot{X}) &= \lambda_1, \\ \frac{d}{dt} (m\dot{Y}) &= \lambda_2, \\ \frac{d}{dt} \left[(mR^2 \sin^2 \theta + I_T) \dot{\theta} \right] - mR^2 \dot{\theta}^2 \sin \theta \cos \theta + I_A (\dot{\psi} - \dot{\phi} \sin \theta) \dot{\phi} \cos \theta \\ &\quad + I_T \dot{\phi}^2 \sin \theta \cos \theta - mgR \sin \theta = \lambda_1 R \cos \theta \sin \phi - \lambda_2 R \cos \theta \cos \phi, \\ \frac{d}{dt} \left[-I_A (\dot{\psi} - \dot{\phi} \sin \theta) \sin \theta + I_T \dot{\phi} \cos^2 \theta \right] &= \lambda_1 R \sin \theta \cos \phi + \lambda_2 R \sin \theta \sin \phi, \\ \frac{d}{dt} \left[I_A (\dot{\psi} - \dot{\phi} \sin \theta) \right] &= -\lambda_1 R \cos \phi - \lambda_2 R \sin \phi. \end{aligned} \quad (9)$$

Equations (9) are combined with the modified constraint relations, that is, from equations (1)

$$\begin{aligned} \dot{X} &= R(\cos \phi) \dot{\psi} - R(\sin \theta \cos \phi) \dot{\phi} - R(\cos \theta \sin \phi) \dot{\theta}, \\ \dot{Y} &= R(\sin \phi) \dot{\psi} - R(\sin \theta \sin \phi) \dot{\phi} + R(\cos \theta \cos \phi) \dot{\theta} \end{aligned} \quad (10)$$

to give a sufficient number of differential equations to solve for the variables X , Y , θ , ϕ , ψ , λ_1 , λ_2 , which would constitute the full solution of the problem.

THE STEADY MOTION OF THE DISK ROLLING IN A CIRCLE

The solution to all nontrivial differential equations is fundamentally by knowing what to guess. The seven differential equations of motion here, equations (9) and (10), are certainly formidable and not trivial to solve in general. A full solution is not offered. However, after watching a coin roll on a horizontal surface, it seems intuitively obvious that a possible steady motion might be the disk rolling in a circle. Therefore, our guess is that the mass center moves in a circle described by

$$\begin{aligned} X &= \rho \sin \omega t, \\ Y &= -\rho \cos \omega t. \end{aligned} \quad (11)$$

The conditions for steady motion are constant velocities and constant values of certain coordinates. They are found to be

$$\begin{aligned} \dot{\phi} &= \omega, \\ \dot{\psi} &= \frac{v}{R} + \dot{\phi} \sin \theta = \frac{\omega \rho}{R} + \omega \sin \theta_0, \\ \dot{\theta} &= 0, \quad \text{or} \quad \theta = \theta_0. \end{aligned} \quad (12)$$

Substituting equations (11) into the first two of the Lagrange's equations (9) leads to the implied values of the Lagrangian multipliers as

$$\begin{aligned} m\ddot{X} &= -m\rho\omega^2 \sin \omega t = \lambda_1, \\ m\ddot{Y} &= +m\rho\omega^2 \cos \omega t = \lambda_2. \end{aligned} \quad (13)$$

These results, equations (11)–(13), satisfy all seven differential equations of motion if the following condition is maintained:

$$\omega^2 = \frac{mgR \tan \theta_0}{I_A(\rho/R) + m\rho R + I_T \sin \theta_0}. \quad (14)$$

This condition for “balance” turning in a circle is developed in several intermediate texts on the subject, e.g., [1], developed with awkward non-Lagrangian method, so as to at least establish this result.

REDUCTION OF EQUATIONS OF MOTION

Taking the time derivative of the constraint relations, equations (10), and substituting the results into the first two of the Lagrange's equations (9), leads to some quite complicated forms for λ_1 and λ_2 . These forms may be substituted into the remaining three Lagrange's equations and reduced (with much algebra) to

$$\begin{aligned} (I_A + mR^2) \frac{d}{dt} (\dot{\psi} - \dot{\phi} \sin \theta) - mR^2 \dot{\theta} \dot{\phi} \cos \theta &= 0, \\ I_A (\dot{\psi} - \dot{\phi} \sin \theta) \dot{\theta} - I_T \ddot{\phi} \cos \theta + 2I_T \dot{\phi} \dot{\theta} \sin \theta &= 0, \quad (15) \\ (I_T + mR^2) \ddot{\theta} + (I_A + mR^2) (\dot{\psi} - \dot{\phi} \sin \theta) \dot{\phi} \cos \theta &+ I_T \dot{\phi}^2 \sin \theta \cos \theta - mgR \sin \theta = 0. \end{aligned}$$

Through this process λ_1 , λ_2 , X , and Y are eliminated, and equations (15) constitute a new set of general equations of motion. Equations (15) constitute a new set of general equations of motion. A similar set of three equations was developed by Pars [2]. However, he did not go further with his analysis. Herein these equations will be analyzed to establish the character of small oscillations about the foregoing steady motion in a circle. They will serve to indicate the minimum rolling speed for stability of the steady motion, as well.

SMALL OSCILLATIONS ABOUT THE STEADY MOTION, ROLLING IN A CIRCLE

In order to analyze small oscillations about the steady motion, the values of the variables for the steady motion itself are denoted with a subscript “0”. Further, small changes in those variables about the steady are denote with an “over-bar”. Defining and tabulating, the new variables gives

$$\begin{aligned} \theta &= \theta_0 + \bar{\theta}, \\ \dot{\phi} &= \omega_0 + \bar{\omega}, \\ \dot{\psi} - \dot{\phi} \sin \theta &= \Omega_0 + \bar{\Omega}, \end{aligned} \quad (16a)$$

where kinematics gives the relation

$$\Omega_0 = \omega_0 \frac{\rho}{R}. \quad (16b)$$

Differentiating these forms leads to

$$\begin{aligned} \dot{\theta} &= \dot{\bar{\theta}}, \\ \ddot{\phi} &= \dot{\bar{\omega}}, \\ \frac{d}{dt} (\dot{\psi} - \dot{\phi} \sin \theta) &= \dot{\bar{\Omega}}. \end{aligned} \quad (17)$$

Upon substitution of these forms, equations (16) and (17), into the first two of the reduced equations of motion, equations (15), and neglecting higher-order terms and further integrating leads to

$$\begin{aligned}\bar{\Omega} &= \frac{mR^2}{I_A + mR^2} \omega_0 (\cos \theta_0) \bar{\theta}, \\ \bar{\omega} &= \frac{I_A \Omega_0 + 2I_T \omega_0 \sin \theta_0}{I_T \cos \theta_0} \bar{\theta}.\end{aligned}\quad (18)$$

Equations (18) show that the variables for small oscillations are linearly coupled and in phase with each other.

The third reduced equation of motion from equations (15) becomes

$$\begin{aligned}(I_T + mR^2) \ddot{\bar{\theta}} + (I_A + mR^2) \left(\frac{\Omega_0 \omega_0 \cos \theta_0 + \omega_0 (\cos \theta_0) \bar{\Omega}}{+ \Omega_0 (\cos \theta_0) \bar{\omega} + \Omega_0 \omega_0 (\sin \theta_0) \bar{\theta}} \right) \\ + \frac{I_T}{2} (\omega_0^2 \sin 2\theta_0 + 2\omega_0 (\sin 2\theta_0) \bar{\omega} + 2\omega_0^2 (\cos 2\theta_0) \bar{\theta}) \\ - mgR \sin \theta_0 - mgR (\cos \theta_0) \bar{\theta} = 0.\end{aligned}\quad (19)$$

Setting the “bar” terms equal to zero returns the usual condition with steady motion (identical to equation (14)) which is

$$(I_A + mR^2) \Omega_0 \omega_0 + I_T \omega_0^2 \sin \theta_0 - mgR \tan \theta_0 = 0. \quad (20)$$

If this condition, equation (20), is used in equation (19), the form of the final result is

$$\ddot{\bar{\theta}} + p^2 \bar{\theta} = 0, \quad (21)$$

where the coefficient, p , which is equal to the circular frequency is

$$p^2 = \frac{mgR}{I_T + mR^2} \cdot \left[\frac{\tan \theta_0}{(I_A + mR^2) (\rho/R) + I_T \sin \theta_0} \left[\begin{aligned} & \frac{mR^2 \cos^2 \theta_0}{+ (I_A + mR^2) \frac{\rho}{R} \left(\frac{I_A \rho}{I_T R} + \sin \theta_0 \right)} \\ & + 2 \left(\frac{I_A \rho}{R} + 2I_T \sin \theta_0 \right) + I_T \cos \theta_0 \end{aligned} \right] \right] - \cos \theta_0 \quad (22)$$

The motion will be stable as long as the preceding bracket, in equation (22) is positive. For motion in a circle of radius, ρ , the sign depends on the magnitude of the rolling angular velocity Ω_0 . That angular velocity is dictated by the condition for steady motion, restated from equation (20) or equation (14) as

$$\Omega_0 = \omega_0 \frac{\rho}{R} = \frac{\rho}{R} \sqrt{\frac{mgR \tan \theta_0}{I_A (\rho/R) + m\rho R + I_T \sin \theta_0}}. \quad (23)$$

For the special case of a wheel rolling in a straight line, in a vertical plane, ($\theta_0 = 0$, $\rho = \infty$), equation (23) gives the result

$$\Omega_{0\text{CR}} = \sqrt{\left(\frac{I_T}{I_A} \right) \frac{mgR}{I_A + mR^2}} \quad (\text{for stable motion}). \quad (24)$$

For a disk with uniformly distributed mass the moments of inertia are

$$I_A = 2I_T = \frac{mR^2}{2}. \quad (25)$$

Substitution into equation (24) results in the well-known form [1]

$$\Omega_{0\text{CR}} = \sqrt{\frac{g}{3R}} \quad (26)$$

for rolling stability with no slip. For a hoop the corresponding result is

$$\Omega_{0\text{CR}} = \sqrt{\frac{g}{4R}}, \quad (27)$$

which is also supported by [1] and other sources.

APPROXIMATE SOLUTION FOR STABILITY FOR THE WHEEL ROLLING IN A CIRCLE OF LARGE RADIUS

Consider a large radius of the circle of motion as relative to the radius of the disk, $\rho \gg R$. The previous general result for stability, equation (22), leading to a positive coefficient is $p^2 = C\{\} \geq 0$. With the large radius assumption the resulting bracket can be expressed approximately as

$$\{\} \cong \tan \theta_0 \frac{I_A \rho}{I_T R} - \cos \theta_0 \geq 0. \quad (28)$$

This result may be rearranged as

$$\frac{\rho}{R} \geq \frac{\cos \theta_0}{\tan \theta_0} \cdot \frac{I_T}{I_A},$$

the condition for steady motion can be written as

$$\Omega_0 \cong \sqrt{\frac{mgR \tan \theta_0}{I_A + mR^2}} \cdot \sqrt{\frac{\rho}{R}}. \quad (29)$$

Combining these approximate results, equations (28) and (29) leads to

$$\Omega_{0 \text{ CR}} \cong \sqrt{\frac{mgR \cos \theta_0}{I_A + mR^2}} \cdot \sqrt{\frac{I_T}{I_A}}. \quad (30)$$

For the disk with uniform distribution of mass, equation (30) with (25) leads to

$$\Omega_{0 \text{ CR}} = \sqrt{\frac{g \cos \theta_0}{3R}}, \quad (31)$$

where as, for the hoop it is

$$\Omega_{0 \text{ CR}} = \sqrt{\frac{g \cos \theta_0}{4R}}. \quad (32)$$

Both of these results differ from the previous results equations (26) and (27) by the cosine of the tilt angle under the root sign. Now cosine is maximum for zero degrees or no tilt corresponding to straight line motion. Therefore, it is noted that the critical rolling speed for stability is actually slower for steady motion in a circle.

REFERENCES

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